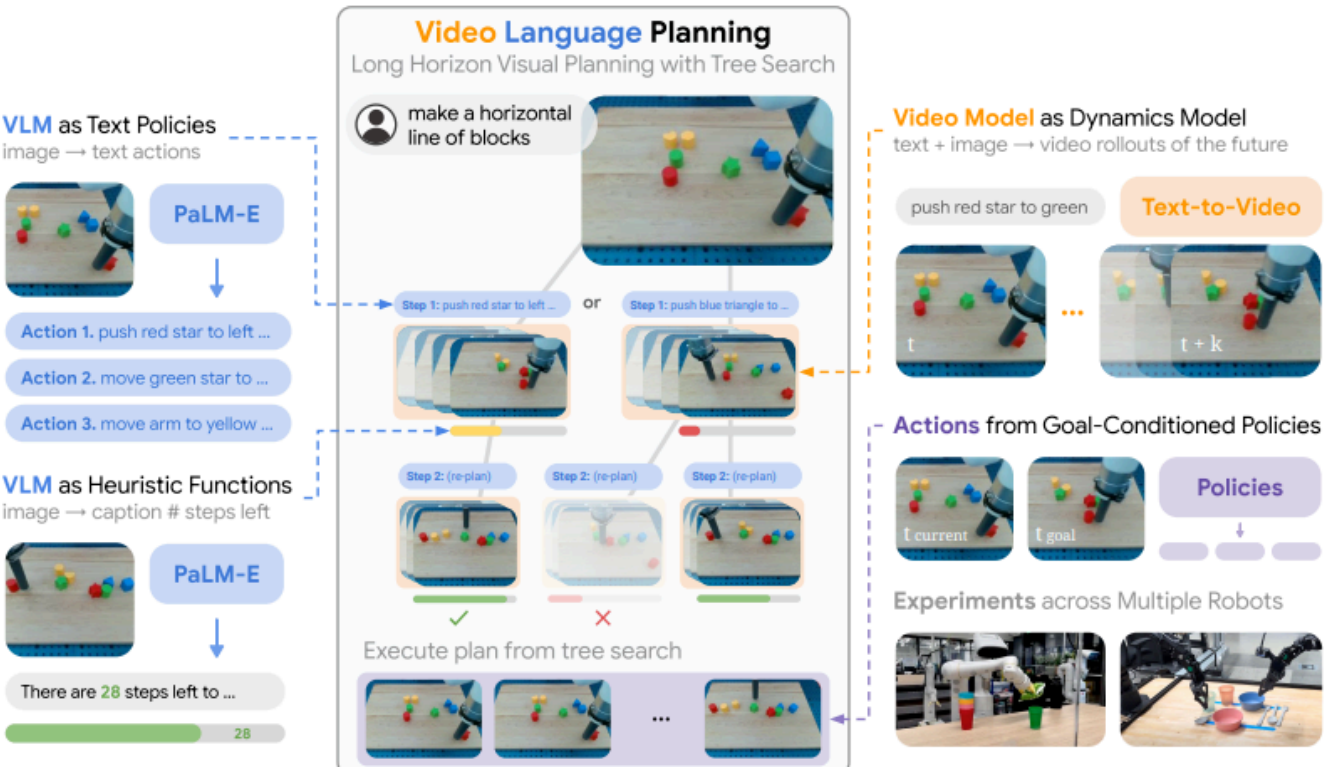
# VIDEO LANGUAGE PLANNING

> #Planning #EmbodiedAI




- What problems does this framework solve?
  - LLMs have powerful inference capabilities, and they can understand and reason abstract planning very well.
  - However, LLMs do not have solid knowledge of the physical world. That is to say, though it knows some basic physics laws, which allow them to generate short videos, it is still hard for them to generate long videos and do physical planning.
- What does this framework do?
  - This framework contains 3 modules:
    - $\pi_{VLM}$: Give the action of the next step. Inputs: Visual signal $x$, NL description of the goal $g$; Outputs: Action $a$.
    - $f_{VM}$: Simulate the physical movement. Inputs: Visual signal $x$, action $a$; Outputs: Video $x_{1:S}$.
    - $H_{VLM}$: Heuristic function. Inputs: A frame $x_n$ and final goal $x^*$; Outputs: A score that measures the difference between the frame and the final goal.
  - Planning Algorithm: Forward Tree Search
    - Start with $B$ beams that share the same origin: the current image.
    - For each beam, use $\pi_{VLM}$ to sample $A$ candidate actions (NL). For each action, generate $D$ videos to simulate the physical changes by $f_{VM}$.
    - Calculate the scores of the videos with $H_{VLM}$. ONLY keep the video with the highest score and concactinate the video to the beam it belongs to. Delete other videos.

        - For every 5 steps, compare the score beam-wisely. Replace the content of the beam with the lowest score with the beam with the highest score.
        - If there is a single-step video that contains anti-physical elements (which means the heuristic will be abnormal), a threshold will be set to filter out such videos.
        - Repeat until reach the given planning length.
    - Inference:
        - Goal-Conditioned Policy: We use a policy model $\pi_{control}$ to receive a set of frames from a **short** sequence of video, compare their differences, and output the low level action that is able to *eliminate the differences in the sequence.*
        - Replanning: Run a few steps of planning each time, and only the 1st few steps will be executed. Afterwards, the new image will be shot, and the planning will be repeated. This is to avoid accumulation of errors
- Experiment Results
    - Significantly improved success rates for long-term tasks: Compared to directly fine-tuned large models (such as PaLM-E and RT-2) or end-to-end video generation models (such as UniPi), plans generated by VLP exhibit superior physical consistency, resulting in a significant lead in execution success rates for long-horizon tasks.
    - Scaling: VLP offers exceptional scalability. As the computational budget increases (e.g., by increasing the branching factors $A$ or $D$ in tree search, or extending the planning horizon), the quality of the generated video plans improves, and task execution accuracy rises accordingly.
    - Excellent Generalization Capabilities: Thanks to the pre-training of the base model on massive amounts of internet data, VLP demonstrates exceptional generalization capabilities. It can not only adapt to new objects and lighting conditions, but also generalize across tasks to entirely new instruction scenarios not seen in the training data.
- Work Cited
    - Paper
    - Chat History
    - NotebookLM